

Risk & Hedging

Lecture Notes

Topics: Risk & Hedging · Forward Contracts · European & American Options · Black-Scholes Model

Risk and Hedging

Risk

The chance that our *actual* outcome differs from our *expected* outcome. *This is direction agnostic.* Example: risk of injury or illness → medical bills, insufficient medical care.

Hedge

A position taken specifically to **offset or reduce** an existing risk exposure. Example: buying health insurance. This doesn't eliminate the risk of injury, but it reduces the risk of high medical bills and insufficient medical care.

Delta Airlines Jet Fuel Hedge (Nov/Dec 2025 → May 2026)

Situation: In late 2025, Delta is planning to run flights during the busy summer season in May 2026. They already know:

- **Relatively Certain:** quantity of jet fuel they will need to purchase in May
- **Uncertain:** the *price* of jet fuel at the time of purchase next May

Risk: The price of jet fuel skyrockets before Delta needs to buy it for summer operations. (e.g. as a result of war).

Possible Action (Hedge): In November/December 2025, Delta could enter into a contract to buy a specified amount of fuel at a fixed price fuel in May 2026. *The supplier gets to lock in a sale now, Delta gets to lock in a purchase now. Removes uncertainty for both parties. This is a **forward contract**.*

Mechanics:

- In November/December 2025, Delta enters a contract agreeing to *buy* jet fuel at a set price for delivery in May 2026.
- Example: They lock in \$3.00/gal for May. When May arrives, the spot market is trading around \$3.90/gal.
- Result: Delta buys at \$0.90/gal *below the spot market price* thanks to the contract.

Forward Contract

Forward Contract

An **agreement to buy** (or sell) an asset at a specified price on a specified future date. Key feature: it *locks in* both the upside and the downside.

Note

Delta is protected against the risk of a price increase, but if the price decreases (e.g. to \$2.00/gal), Delta would still be obligated to buy at the contracted price (e.g. \$3.00/gal). A forward contract eliminates both the upside and the downside risk. **There is a way to protect against the downside while *preserving* the upside**: this is where **options** come in.

Options: Definitions and Payoffs

Option

A contract that gives the holder the **right, but not the obligation**, to buy or sell a particular asset at a particular price on (or before) a specified date.

Call vs. Put

- **Call option**: the right to **buy** the underlying asset at the strike price.
- **Put option**: the right to **sell** the underlying asset at the strike price.

European vs. American Options

- **European option**: can only be exercised *on* the expiration date T .
- **American option**: can be exercised *at any time* up to and including the expiration date.

Key Notation

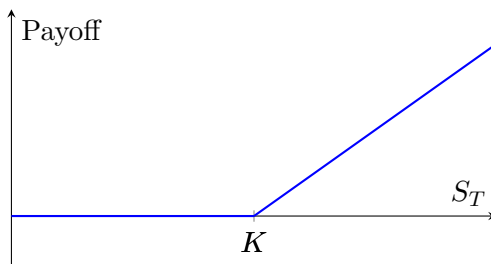
Symbol	Meaning
K	Strike price (the agreed purchase/sale price)
T	Expiration date
S_T	Price of the underlying asset at time T
S_0	Price of the underlying asset today ($t = 0$)
C	Option premium (the price paid to buy the option)

Call Option Payoff

$$\text{Payoff at } T = \max(S_T - K, 0)$$

- If $S_T > K$: the option is **in-the-money**; you exercise and earn $S_T - K$.
- If $S_T \leq K$: the option expires **worthless**; you lose only the premium C .
- The maximum *loss* to the buyer is the premium C paid at $t = 0$.

Payoff Diagram — Long Call



Gain/Loss Tables (Numerical Example)

Suppose the current asset price is $S_0 = \$10$ and you buy a call option with strike $K = \$10$ and premium $C = \$1$. Let's compare the final gain/loss for buying the asset outright vs. buying the call option under different future asset prices.

S_T	Buy Stock G/L	Buy Call G/L
\$5	-\$5	-\$1
\$10	\$0	-\$1
\$11	+\$1	\$0
\$15	+\$5	+\$4

Note. The gain/loss from buying the stock is simply $S_T - S_0$. For the call, the gain/loss is $-C$ if $S_T \leq K$ (option expires worthless, loss equals the premium), and $S_T - K - C$ if $S_T > K$ (option is exercised). In the table, $S_0 = K = \$10$ and $C = \$1$.

Note

Hedging vs. Speculation with Options

We initially set up the option idea as a *hedge*: offsetting an existing exposure (like Delta's fuel cost). But options are also powerful **speculation** tools:

- You gain exposure to an asset's price movement.
- Only a small capital outlay (the premium C) is required.
- Loss is capped at the premium; gains are theoretically unlimited (for calls).

Black-Scholes Model

Black-Scholes Model

A mathematical model used to **price a European call option** on a non-dividend-paying underlying asset. The formula estimates the price the option *should* trade at given observable market inputs and an estimated volatility σ .

The Black-Scholes Formula

$$C = S_0 \cdot \Phi(d_1) - Ke^{-rT} \cdot \Phi(d_2)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

and $\Phi(\cdot)$ is the standard normal CDF.

Intuition: Two Components

(A) $S_0 \cdot \Phi(d_1)$ — Present Value of the Asset Received

- Represents the **present value of the asset you receive** (the underlying S_0), weighted by the probability that the option finishes in the money, adjusted for the magnitude of the payoff.

(B) $Ke^{-rT} \cdot \Phi(d_2)$ — Present Value of the Strike Price Paid

- Represents the **present value of what you pay** (the strike price K), weighted by the probability that the option is exercised.

$$C = \underbrace{S_0 \cdot \Phi(d_1)}_{\text{Expected asset value received (PV)}} - \underbrace{Ke^{-rT} \cdot \Phi(d_2)}_{\text{Expected strike paid (PV)}}$$

Note

All-or-Nothing intuition: $\Phi(d_2)$ answers “Will the option finish in-the-money at all?” while $\Phi(d_1)$ answers “How far in-the-money is it expected to finish?” The Black-Scholes price is the *probability-weighted, present-value difference* between what you get and what you pay.